Journal of Statistical Software

MMMMMM YYYY, Volume VV, Issue II. doi: 10.18637/jss.v000.i00

exdqlm: An R Package for Estimation and Analysis of Flexible Dynamic Quantile Linear Models

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Abstract

We present the R package **exdqlm** as a tool for dynamic quantile regression. The main focus of the package is to provide a framework for Bayesian inference and forecasting of flexible dynamic quantile linear models (exDQLMs). exDQLMs utilize a new family of error distributions for parametric quantile regression, the extended asymmetric Laplace (exAL), a generalization of the asymmetric Laplace (AL) which is commonly used in parametric quantile regression. Estimation of a dynamic quantile linear model, which utilizes the AL, is included in the package as a special case. Non-time-varying quantile regression models are also a special case of our methods. Further, routines for estimation of a nonlinear relationship between the response and a given input variable at a specified quantile via a transfer function model are available. The software provides exact posterior inference via Markov chain Monte Carlo (MCMC) and fast approximate posterior inference via Laplace delta variational Bayes (LDVB). While MCMC provides sample-based exploration of the posterior distribution, the approximations obtained using the LDVB algorithm enable fast inference for long time series at a fraction of the memory and computational costs of the MCMC. A routine for forecasting the dynamic quantile is available in the package, as well as several quantitative and visual diagnostics for model evaluation. We illustrate the implementation of the functions and algorithms in the **exdqlm** package with step-by-step analyses of several real and simulated data sets.

Keywords: dynamic quantile regression, dynamic models, asymmetric Laplace, variational Bayes, coordinate ascent variational inference, quantile regression.

1. Introduction

Quantile regression has gained popularity in the non-time-varying setting as a robust alternative to mean-centric approaches, but time-varying methods and software implementations remain limited. In a Bayesian setting, parametric quantile regression models are almost exclusively based on the asymmetric Laplace (AL) likelihood because it corresponds to the check-loss function minimized in traditional quantile regression problems (Yu and Moyeed 2001; Koenker 2005). A mixture representation of the AL provides computational advantages for posterior inference, which has contributed to its popularity, but the AL is known to be restrictive in terms of flexibility. Very general error distributions for quantile regression have been considered extensively in the Bayesian non-parametric literature (Kottas and Krnjajić 2009; Reich, Bondell, and Wang 2009; Taddy and Kottas 2010), however the computational cost of these methods does not scale well to dynamic models for long time series. Yan and Kottas (2017) present a parametric extension of the AL, the extended asymmetric Laplace (exAL), which overcomes these restrictive aspects while preserving the computational advantages of the AL. Barata, Prado, and Sansó (2022) use this family of error distributions to develop a flexible dynamic quantile linear model (exDQLM), which is the model implemented in the R package **exdqlm**.

The purpose of **exdqlm** is to provide a framework for Bayesian inference, forecasting, and model assessment for a single dynamic quantile. The package implements an MCMC algorithm for exact posterior inference and an LDVB algorithm for fast approximate posterior inference. It also includes routines for transfer function exDQLMs, which provide a straightforward way to model nonlinear relationships between a response and an input variable at a specified quantile, along with MCMC and LDVB routines for static quantile regression under the exAL error distribution. Forecasting and several visual and quantitative diagnostics are also available. A legacy ISVB routine is retained for backward compatibility, although in this paper we focus on LDVB as the primary approximation. The package is protected under the MIT License and is available from the Comprehensive R Archive Network (CRAN) at `https://CRAN.R-project.org/package=exdqlm`.

Well-developed software implementations of non-time-varying quantile regression models under the AL are available in the Bayesian R package **bayesQR** (Benoit, Al-Hamzawi, Yu, and Van den Poel 2011) and, equivalently, a classical version based on the check-loss function (Koenker 2005) is available in the package **quantreg** (Koenker 2021). Since the AL is a special case of the exAL, the methods used in **bayesQR** are also a special case of the framework implemented in **exdqlm**. There are also many established packages for dynamic modeling, including **dlm** (Petris and Gilks 2018) and **dynr** (Ou, Hunter, Chow, Ji, Chen, Hung, Lee, Li, and Park 2021), but implementations for dynamic quantile modeling remain much more limited. To our knowledge, the package **quantreg** includes a routine that allows for a time-evolving structure in the regression coefficients, which is itself a special case of the routines in our package. In the remainder of this paper we introduce the package as follows. In Section 2, we describe the modeling framework, including the algorithms used for posterior inference and our methods for including nonlinear relationships. We also discuss static quantile regression as a special case of the framework. In Section 3 we detail the model diagnostics and forecasting method implemented in the package. Then, in Section 4, we walk through four step-by-step examples on how to use the package. Finally, in Section 5, we present a summary of the package capabilities.

2. Extended dynamic quantile linear models

Consider a set of scalar time-evolving responses, y_t, for times $t = 1, \ldots, T$. For each t, an extended dynamic quantile linear model (exDQLM) for inference on a single p_0-th quantile is defined by

$$\text{Observation equation:} \quad y_t = \mathbf{F}_t' \boldsymbol{\theta}_t + \epsilon_t, \qquad \epsilon_t \sim \text{exAL}_{p_0}(0, \sigma, \gamma) \tag{1}$$

$$\text{System equation:} \quad \boldsymbol{\theta}_t = \mathbf{G}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\omega}_t, \qquad \boldsymbol{\omega}_t \sim \text{N}(\mathbf{0}, \mathbf{W}_t). \tag{2}$$

Here $\mathbf{F}_t$ is the $q \times 1$ regression vector of the covariates corresponding to the $q \times 1$ parameter vector $\boldsymbol{\theta}_t$ at time t, $\mathbf{G}_t$ is the $q \times q$-dimensional evolution matrix defining the structure of the parameter vector evolution in time, and the normal distribution according to which the system error $\boldsymbol{\omega}_t$ evolves is q-variate. The observational errors ϵ_t of the exDQLM follow an exAL distribution for fixed quantile p_0, whose density is denoted as $\text{exAL}_{p_0}(\cdot)$, and it is such that $\int_{-\infty}^{0} \text{exAL}_{p_0}(\epsilon_t|0, \sigma, \gamma) \text{d}\epsilon_t = p_0$. Thus, the observation equation in Equation (1) implies that $\mathbf{F}_t' \boldsymbol{\theta}_t$ corresponds to the p_0-quantile of y_t, even if this is not made explicit in the notation above. Table 1 shows the function parameters that define the structure of the dynamic quantile, specifically $\mathbf{F}_t$ and $\mathbf{G}_t$ for $t = 1, \ldots, T$.

The exAL distribution is constructed from an extension of the location-scale mixture representation of the AL (Kozumi and Kobayashi 2011), details of which can be found in Yan and Kottas (2017). This results in a mixture representation of $\text{exAL}_{p_0}(0, \sigma, \gamma)$ as follows:

$$\int \int_{\mathbb{R}^+ \times \mathbb{R}^+} \text{N}(y_t | C(p, \gamma)\sigma|\gamma| s_t + A(p) v_t, \sigma B(p) v_t) \text{Exp}(v_t|\sigma) \text{N}^+(s_t|0, 1) \text{d}v_t \text{d}s_t. \tag{3}$$

Here, $p = p(p_0, \gamma) = I(\gamma < 0) + \{[p_0 - I(\gamma < 0)]/g(\gamma)\}$ where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and $\Phi(\cdot)$ denotes the standard normal CDF. $A(p)$, $B(p)$, $C(p, \gamma)$ are the functions of p and γ: $A(p) = \frac{1-2p}{p(1-p)}$, $B(p) = \frac{2}{p(1-p)}$, $C(p, \gamma) = [I(\gamma > 0) - p]^{-1}$. Lastly, $\text{Exp}(v|\sigma)$ denotes the exponential distribution with mean σ, and $\text{N}^+(s_t|0, 1)$ denotes a normal distribution truncated to the positive reals with mean 0 and variance 1. This mixture representation of the exAL can be exploited to rewrite the exDQLM as the following hierarchical model for $t = 1, \ldots, T$:

$$y_t | \boldsymbol{\theta}_t, \sigma, \gamma, v_t, s_t \sim \text{N}(y_t | \mathbf{F}_t' \boldsymbol{\theta}_t + C(p, \gamma)\sigma|\gamma| s_t + A(p) v_t, \sigma B(p) v_t) \tag{4}$$

$$v_t, s_t | \sigma \sim \text{Exp}(v_t|\sigma) N^+(s_t|0, 1) \tag{5}$$

$$\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t-1}, \mathbf{W}_t \sim \text{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t). \tag{6}$$

This hierarchical form is leveraged for the posterior inference implemented within the **exdqlm** package.

2.1. Prior specification

Under a Bayesian formulation and given p_0, we specify priors for the initial state vector $\boldsymbol{\theta}_0$, scale parameter σ and skewness parameter γ. For $\boldsymbol{\theta}_0$, we assume a q-variate conjugate normal prior $\boldsymbol{\theta}_0 \sim \text{N}(\mathbf{m}_0, \mathbf{C}_0)$. For σ, we assume a conjugate inverse-gamma prior denoted as $\text{IG}(a_\sigma, b_\sigma)$. In many cases a strong prior on σ is necessary to guarantee fast posterior convergence and estimation. This is illustrated in the examples of Section 4. The parameter γ has bounded support over the interval (L, U) where L is the negative root of $g(\gamma) = 1 - p_0$ and U is the positive root of $g(\gamma) = p_0$ (Yan and Kottas 2017). In contrast to

Input list	`model`				`PriorSigma`		`PriorGamma`		
Symbol	$\mathbf{F}_{1:T}$	$\mathbf{G}_{1:T}$	$\mathbf{m}_0$	$\mathbf{C}_0$	a_σ	b_σ	m_γ	s_γ	ν_γ
List component	`FF`	`GG`	`m0`	`C0`	`a_sig`	`b_sig`	`m_gam`	`s_gam`	`df_gam`
Default	-	-	$\mathbf{0}_q$	$1e3\mathbf{I}_q$	2.1	1.1	0	1	1

Table 1: Translation from mathematical symbols to parameters used by **exdqlm** functions, as well as default values where applicable. $\mathbf{0}_q$ denotes a q-dimensional vector of zeros, and $\mathbf{I}_q$ denotes the q-dimensional idenity matrix.

the flat prior suggested by Yan and Kottas (2017), we find that using a proper prior on the skewness parameter promotes reliable posterior inference. Thus, we place a Student-t distribution truncated to the interval (L, U) as the prior for γ, i.e. $\gamma \sim \mathrm{t}_{(L,U)}(m_\gamma, s_\gamma)$ with ν_γ degrees of freedom. The prior parameters used in the **exdqlm** functions, their corresponding mathematical symbols, and default values are shown in Table 1.

2.2. Posterior estimation

The hierarchical representation of the exDQLM facilitates posterior simulation using Markov Chain Monte Carlo (MCMC) algorithms. This is implemented in the function `exdqlmMCMC()`. Gibbs sampling steps are used to update the sets of latent variables s_t and v_t for $t = 1, \ldots, T$. The dynamic regression coefficients are sampled using a forward filtering backwards sampling algorithm (Carter and Kohn 1994; Frühwirth-Schnatter 1994). The scale and skewness parameters are then updated separately from the latent-state blocks. By default, the package uses a slice-sampling update for the (σ, γ) block, drawing σ from its full conditional distribution and updating γ with a bounded univariate slice sampler. Alternatively, the package also provides joint random-walk Metropolis-Hastings (MH) kernels on transformed coordinates, including a Laplace-informed proposal covariance that can be refreshed during burn-in. When desired, the chain can be warm-started from either the LDVB or legacy ISVB approximation; this initialization is separate from the subsequent MCMC proposal kernel. An example of this functionality can be found in Section 4.1. We perform a total of $N_{BURN} + N_{MCMC}$ MCMC iterations, discarding the first N_{BURN} and retaining the remaining N_{MCMC} posterior samples. The object created by `exdqlmMCMC()` includes posterior samples of all parameters $(\boldsymbol{\theta}_{1:T}, v_{1:T}, s_{1:T}, \gamma, \sigma)$, filtered and smoothed summaries for the state vector $\boldsymbol{\theta}_{1:T}$, and diagnostics for the proposal kernel used in the (σ, γ) update.

For purposes of model selection or large time series data sets, posterior sampling with the MCMC algorithm can be computationally demanding. To this end, we include in the package a Laplace delta variational Bayes (LDVB) routine, `exdqlmLDVB()`, for fast and accurate approximate posterior estimation. The variational Bayes (VB) framework approximates the posterior distribution with partitioned variational distributions (Ostwald, Kirilina, Starke, and Blankenburg 2014; Beal 2003). Under a mean-field factorization, this induces a blockwise coordinate-ascent variational inference scheme. For the exDQLM, the blocks associated with the dynamic coefficients and latent variables admit recognizable closed-form updates, while the joint variational distribution of σ and γ is non-conjugate. Following the non-conjugate VB strategy of Wang and Blei (2013), we approximate this block with a Laplace approximation on transformed coordinates and then use a second-order Delta method to evaluate the expectations of functions of (σ, γ) needed by the remaining variational updates. Thus,

exdqlm function	`exdqlmMCMC()`				`exdqlmLDVB()`	
Symbol	$\mathcal{K}_{\sigma,\gamma}$	Σ_{MH}	N_{BURN}	N_{MCMC}	ϵ_{tol}	N_{SAMP}
Input	`mh.proposal`	`Sig.mh`	`n.burn`	`n.mcmc`	`tol`	`n.samp`
Default	`"slice"`	$0.05\mathbf{I}_2$	2000	1500	0.1	200

Table 2: Translation from mathematical symbols to parameters used in functions `exdqlmMCMC()` and `exdqlmLDVB()`, as well as default values.

the mean-field factorization induces the blockwise CAVI updates, while the non-conjugate (σ, γ) block is handled through the Laplace-Delta approximation. The dynamic regression coefficients are updated within each iteration using a forward filtering backwards smoothing algorithm. Convergence is assessed jointly through the state, scale, skewness, and ELBO diagnostics. After the algorithm reaches convergence with tolerance ϵ_{tol}, N_{SAMP} samples are drawn from the variational distributions, resulting in an approximate sample of the posterior distribution. The output of `exdqlmLDVB()` includes the approximate posterior samples as well as the parameters of the variational distributions for all parameters $(\boldsymbol{\theta}_{1:T}, v_{1:T}, s_{1:T}, \gamma, \sigma)$. For the (γ, σ) block, the return also contains the transformed mode, approximate covariance matrix, derived moments, and ELBO terms associated with the Laplace-Delta approximation. The package also retains a legacy importance-sampling VB routine, `exdqlmISVB()`, but in this paper we restrict attention to MCMC and LDVB as the primary inferential tools.

Table 2 shows the input parameters of the fitting functions `exdqlmMCMC` and `exdqlmLDVB`, their mathematical symbols, and their default values.

The argument `Sig.mh` is only used when `mh.proposal` selects a joint random-walk MH kernel; it is not used under the default slice-sampling update.

2.3. Specification of the evolution covariance via discount factors

Both estimation algorithms, `exdqlmMCMC` and `exdqlmLDVB`, utilize discount factors to specify the time-evolving covariance matrices $\mathbf{W}_t$ introduced in Equation (2) (West, Harrison, and Migon 1985). More precisely, we define the evolution variance matrix $\mathbf{W}_t = \frac{1-\delta}{\delta}\mathbf{G}_t\mathbf{C}_{t-1}\mathbf{G}_t'$ where $\mathbf{C}_{t-1}$ denotes the posterior variance of the state vector $\boldsymbol{\theta}_{t-1}$ at time $t-1$ and δ is a discount factor such that $0 < \delta \leq 1$. The discount factor δ can be specified by the user via function parameter `df`. Selection of discount factors is typically done by optimizing a model-checking criterion. We suggest using the KL divergence or posterior predictive loss criterion discussed in Section 3. A brief example of this method for discount factor selection can be found in Section 4.2.

Discounting by component of the state-space model is also a functionality of our estimation algorithms. Suppose the state vector is comprised of h components $\boldsymbol{\theta}_{it}$ of dimension q_i for $i = 1, \ldots, h$ such that $\boldsymbol{\theta}_t' = (\boldsymbol{\theta}_{1t}', \ldots, \boldsymbol{\theta}_{ht})'$ and $\sum_{i=1}^{h} q_i = q$. If $\mathbf{W}_{it}$ denotes the ith component of the evolution covariance matrix such that $\mathbf{W}_t = \text{blockdiag}\{\mathbf{W}_{1t}, \ldots, \mathbf{W}_{ht}\}$ and δ_i denotes a discount factor for the ith component, we can define the evolution variance matrix by component as $\mathbf{W}_{it} = \frac{1-\delta_i}{\delta_i}\mathbf{G}_{it}\mathbf{C}_{i,t-1}\mathbf{G}_{it}'$. Here $\mathbf{G}_{it}$ and $\mathbf{C}_{i,t-1}$ denote the ith components of $\mathbf{G}_t$ and $\mathbf{C}_{t-1}$, respectively. The function parameters `df` and `dim.df` allow users to specify a set of component discount factors $(\delta_1, \ldots, \delta_h)$ and their dimensions $(q_1, \ldots, q_h)$, respectively, within the state-space model. Examples of this feature can be found in Section 4. For a full review of discount factors including discounting by component, see West *et al.* (1985).

Symbol	X_t	$\tilde{\lambda}$	$(\delta_\zeta, \delta_\psi)$	$\mathbf{m}_0^{tf}$	$\mathbf{C}_0^{tf}$
`transfn_exdqlmLDVB()` input	`X`	`lam`	`tf.df`	`tf.m0`	`tf.C0`

Table 3: Translation from mathematical symbols to parameters used in function `transfn_exdqlmLDVB()`.

2.4. Transfer function model

Quantifying the relationship between an input and a quantile of a response is a non-trivial task, particularly when the relationship is nonlinear. In mean-centric dynamic modeling, transfer functions are a straightforward method to incorporate variables which measure the combined effect of current and past regression effects (West *et al.* 1985). An extension of the transfer function approach to the dynamic quantile modeling setting is implemented within the **exdqlm** package, particularly for the LDVB algorithm.

For $t = 1, \ldots, T$ and univariate input, X_t, a transfer function exDQLM with exponential decay is defined by

$$y_t|\boldsymbol{\theta}_t, \gamma, \sigma \sim \text{exAL}_{p_0}(\mathbf{F}_t'\boldsymbol{\theta}_t + \zeta_t, \sigma, \gamma) \tag{7}$$

$$\boldsymbol{\theta}_t|\boldsymbol{\theta}_{t-1}, \mathbf{W}_t \sim \text{N}(\mathbf{G}_t\boldsymbol{\theta}_{t-1}, \mathbf{W}_t) \tag{8}$$

$$\zeta_t|\zeta_{t-1}, \psi_{t-1}, \omega_t \sim \text{N}(\lambda\zeta_{t-1} + X_t\psi_{t-1}, \omega_t) \tag{9}$$

$$\psi_t|\psi_{t-1}, \nu_t \sim \text{N}(\psi_{t-1}, \nu_t). \tag{10}$$

Here ζ_t captures the effect of the current and past regression effects on the quantile at time t. Alternatively, the parameter ψ_t determines the instantaneous effect X_t has on the quantile. The parameter $\lambda \in [0, 1]$ represents the memory of the regression effect up to time t. This effect decays at an exponential rate, reducing by a factor of λ at every time step. More explicitly, the effect of X_t on the quantile at time $t + k$ is $\lambda^k\psi_{t-1}X_t$. Thus, the series $k_t \geq \frac{\log(\epsilon)-\log(|\psi_{t-1}X_t|)}{\log(\lambda)}$ represents a lower bound for the number of time steps until the effect of X_t is less than or equal to a fixed ϵ. The median value of k_t corresponding to $\epsilon = 1\text{e}{-3}$ is included as an output of our transfer function routine.

Conditional on a fixed λ, say $\tilde{\lambda}$, we augment the LDVB algorithm to incorporate the transfer function structure by simply replacing $\mathbf{F}_t, \boldsymbol{\theta}_t, \mathbf{G}_t$, and $\mathbf{W}_t$ in Equations (1)-(2) with $\tilde{\mathbf{F}}_t' = (\mathbf{F}_t', 1, 0)$, $\tilde{\boldsymbol{\theta}}_t' = (\boldsymbol{\theta}_t', \zeta_t, \psi_t)$, $\tilde{\mathbf{G}}_t = \text{blockdiag}\left\{\mathbf{G}_t, \begin{pmatrix} \tilde{\lambda} & X_t \\ 0 & 1 \end{pmatrix}\right\}$, and $\tilde{\mathbf{W}}_t = \text{blockdiag}\left\{\mathbf{W}_t, \begin{pmatrix} \omega_t & 0 \\ 0 & \nu_t \end{pmatrix}\right\}$, respectively. This augmentation is implemented in the function `transfn_exdqlmLDVB()`, an extension of the routine found in `exdqlmLDVB()`.

Similar to the specification of $\mathbf{W_t}$ discussed in Section 2.3, discount factors $(\delta_\zeta, \delta_\psi)$ are utilized to specify $\begin{pmatrix} \omega_t & 0 \\ 0 & \nu_t \end{pmatrix}$. A bivariate normal conjugate prior on $(\zeta_0, \psi_0)' \sim N(\mathbf{m}_0^{tf}, \mathbf{C}_0^{tf})$ completes the transfer function model extension. Table 3 shows the additional parameters used in `transfn_exdqlmLDVB()`, as well as their mathematical symbols discussed in this section.

2.5. Special cases, including static quantile regression

The first special case of our exDQLM arises from the fact that the exAL is an extension of the AL (Yan and Kottas 2017). When $\gamma = 0$, the observational error ϵ_t in Equation (1) reduces such that $\epsilon_t \sim \text{exAL}_{p_0}(\epsilon_t|0, \sigma, \gamma) = \text{AL}_{p_0}(\epsilon_t|0, \sigma)$. This special case with observational errors distributed independently from an AL is the dynamic quantile linear model (DQLM)

presented in Gonçalves, Migon, and Bastos (2017). The DQLM can be implemented with our package using the parameter `dqlm.ind = TRUE`.

Non-time-varying quantile regression is also a special case of our methodology. More specifically, when discount factors are set to 1, the evolution covariance $\mathbf{W}_t$ reduces to $\mathbf{0}$ resulting in non-time-varying parameters (West *et al.* 1985). Thus, the model reduces to the static exAL regression methodology of Yan and Kottas (2017). This model can be implemented using our functions `exal_static_mcmc()` or `exal_static_LDVB()`, which implement MCMC and LDVB algorithms, respectively.

The final special case is the intersection of the previous two. A static quantile regression model with observational errors distributed according to an AL corresponds to Bayesian linear quantile regression under the AL working likelihood, first presented in Yu and Moyeed (2001). This methodology, utilized to create the Bayesian package for quantile regression **bayesQR**, can also be implemented with our package using functions `exal_static_mcmc()` or `exal_static_LDVB()` with parameter `dqlm.ind = TRUE`, which acts as a convenience flag fixing $\gamma = 0$.

Examples of these special cases can be found in Section 4. A static quantile regression example specifically can be found in Section 4.4. For further details on the methodology outlined in this section, see Barata *et al.* (2022).

3. Model Diagnostics and Forecasting

To evaluate the quantile inference and predictive performance of the exDQLM, several quantitative and visual model diagnostics are included in the package.

The one-step-ahead predictive distribution function introduced by Rosenblatt (1952) is a useful diagnostic for models in which the marginal forecast distributions are unrecognizable or difficult to compute (Huerta, Jiang, and Tanner 2003; Prado, Molina, and Huerta 2006). For the exDQLM, this distribution is defined as

$$u_t = \Pr(Y_t \leq y_t | y_{1:t-1}, v_{1:T}, s_{1:T}, \sigma, \gamma), \tag{11}$$

with $y_{1:t-1} = y_1, \ldots, y_{t-1}$, $v_{1:T} = (v_1, \ldots, v_T)$ and $s_{1:T} = (s_1, \ldots, s_T)$. Here u_t defines an independent sequence which is uniformly distributed on the interval $(0, 1)$ (Rosenblatt 1952). Conditional on $\{v_{1:T}, s_{1:T}, \sigma, \gamma\}$, the forecast distribution of y_t within the exDQLM is normally distributed. Therefore, conditional on MAP estimates $\{\hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}\}$ we estimate the sequence as

$$\hat{u}_t = \Phi(y_t | y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}) \tag{12}$$

where Φ denotes the normal CDF.

Using this estimated sequence $\{\hat{u}_t\}$, we can assess the model performance in several ways. First, we visually examine the correlation of the estimated sequence via an ACF plot. Second, by transforming the values with a standard normal inverse CDF, we compare their distributional shape to that of a standard normal with a normal QQ-plot. To quantify the divergence of this transformed sequence from normality we compute the KL divergence (Kullback and Leibler 1951), $\mathrm{KL}(h, \phi) = \int_{-\infty}^{\infty} h(x) \log \frac{h(x)}{\phi(x)} \mathrm{d}x$, where h denotes the numerically approximated density of our transformed sample and ϕ is the standard normal density. A smaller

Symbol	$\tilde{t}$	k	$\mathbf{F}_{(\tilde{t}+1):(\tilde{t}+k)}$	$\mathbf{G}_{(\tilde{t}+1):(\tilde{t}+k)}$
`exdqlmForecast()` input	`start.t`	`k`	`fFF`	`fGG`

Table 4: Translation from mathematical symbols to additional parameters used in function `exdqlmForecast()`.

KL divergence suggests a superior model fit. Lastly, the MAP standardized forecast errors,

$$\frac{y_t - \mathsf{E}[y_t|y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma}]}{\sqrt{\mathrm{Var}(y_t|y_{1:t-1}, \hat{v}_{1:T}, \hat{s}_{1:T}, \hat{\sigma}, \hat{\gamma})}} \tag{13}$$

can also be used as an additional graphical aid in examining the distribution at specific time points. These MAP standardized forecast errors are included in the output of the `exdqlmMCMC` and `exdqlmLDVB` functions.

For a final diagnostic, we include Gelfand and Ghosh (1998) posterior predictive loss criterion (pplc) with check loss function $\rho_{p_0}(x) = x[p_0 - I(x < 0)]$. That is,

$$\text{pplc} = \sum_t \mathsf{E}[\rho_{p_0}(y_t^{obs} - y_t^{rep})|y_{1:T}], \tag{14}$$

where the expectation is with respect to the posterior predictive distribution of y_t, $p(y_t^{rep}|y_{1:T})$. This distribution is used to generate a replicate dataset and is the expected value of the observation equation (1) with respect to the posterior distributions of the parameters. Again, a smaller pplc value suggest a superior model fit. Samples from the posterior replicate distributions are included in the output of the `exdqlmMCMC()` and `exdqlmLDVB()` functions.

The function `exdqlmDiagnostics()` implements the model checking criteria discussed in this section. When an output from either `exdqlmMCMC()` and `exdqlmLDVB()` is passed to `exdqlmDiagnostics()`, the resulting object includes: the estimated sequence $\{\hat{u}_t\}$, the KL divergence, pplc, the ordered pairs of the QQ-plot, and autocorrelations by lag. By default, the function also produces the QQ-plot, ACF plot and MAP standard forecast error plot. Examples of the utility of `exdqlmDiagnostics()` can be found in the following Section 4.

3.1. k-step-ahead forecast

For an arbitrary time $\tilde{t}$, the k-step-ahead future marginal distribution of the quantile is

$$\mathbf{F}'_{\tilde{t}+k}\boldsymbol{\theta}_{\tilde{t}+k}|y_1, \ldots, y_{\tilde{t}} \sim \mathrm{N}(\mathbf{F}'_{\tilde{t}+k}a_{\tilde{t}}(k), \mathbf{F}'_{\tilde{t}+k}\mathbf{R}_{\tilde{t}}(k)\mathbf{F}_{\tilde{t}+k}) \tag{15}$$

where $\mathbf{a}_{\tilde{t}}(k) = \mathbf{G}_{\tilde{t}+k}\mathbf{a}_{\tilde{t}}(k-1)$, $\mathbf{R}_{\tilde{t}}(k) = \mathbf{G}_{\tilde{t}+k}\mathbf{R}_{\tilde{t}}(k-1)\mathbf{G}'_{\tilde{t}+k} + \mathbf{W}_{\tilde{t}+k}$, $\mathbf{a}_{\tilde{t}}(0) = \mathbf{m}_{\tilde{t}}$, and $\mathbf{R}_{\tilde{t}}(0) = \mathbf{C}_{\tilde{t}}$, with $\mathbf{m}_{\tilde{t}}$ and $\mathbf{C}_{\tilde{t}}$ denoting the filtered mean and covariance of $\boldsymbol{\theta}_{\tilde{t}}$, respectively. This distribution is implemented in the function `exdqlmForecast` which is compatible with the outputs from both the MCMC and LDVB algorithms. Table 4 shows the input parameters used in the function, as well as their mathematical symbols seen in this section. The output of `exdqlmForecast()` includes the parameters of the forecasted distribution as well as $\mathbf{a}_{\tilde{t}}(k)$ and $\mathbf{R}_{\tilde{t}}(k)$ for the k steps.

Legacy ISVB routines remain available in the package for backward compatibility, but they are omitted from the primary workflow summary in Table 5.

Function	Output	Description
Model construction		
`polytrendMod()`	`exdqlm`	Create n-th order polynomial component.
`seasMod()`	`exdqlm`	Create Fourier form periodic component.
`regMod()`	`exdqlm`	Create standard regression component.
`as.exdqlm()`	`exdqlm`	Manipulates `list` into `exdqlm` object.
Model fitting – Input `exdqlm` objects		
`exdqlmMCMC()`	`exdqlmMCMC`	Estimate the posterior with MCMC algorithm.
`exdqlmLDVB()`	`exdqlmLDVB`	Estimate the posterior with LDVB algorithm.
`transfn_exdqlmLDVB()`	`exdqlmLDVB`	Estimate transfer function exDQLM using LDVB.
Model fit exploration – Input `exdqlmMCMC` or `exdqlmLDVB` objects		
`exdqlmForecast()`	`exdqlmForecast`	Compute k-step-ahead forecast quantiles.
`exdqlmDiagnostics()`	`exdqlmDiagnostic`	Compute & plot model diagnostics.
`exdqlmPlot()`	`list`	Plot posterior summary of dynamic quantile.
`compPlot()`	`list`	Plot posterior summary of component/parameter.

Table 5: Overview of functions available for dynamic quantile regression in **exdqlm**. Generic S3 methods defined for the objects listed in the table include `plot()`, `print()`, and `summary()`.

4. Examples

Table 5 provides an overview of the **exdqlm** functions and the resulting objects. We demonstrate the general workflow and key features of the package through several analyses on real datasets. The first three examples illustrate dynamic workflows, while the fourth illustrates the static exAL special case. The following examples are implemented on a personal computer with an Apple M1 processor. We begin by loading the **exdqlm** package used for all examples.

```
R> library(exdqlm)
```

4.1. Lake Huron

In this example, we consider the Lake Huron time series from the package **datasets**. The data are annual measurements of the level (ft) of Lake Huron from 1875 to 1972, shown in Figure 2. With this example we show how to: build a basic state-space structure, specify a prior on γ, run the MCMC algorithm, plot the estimated quantiles and forecast distributions.

To estimate the dynamic distribution of the data, we consider three quantiles, $p_0 = 0.95, 0.50,$ and 0.05. We begin by creating the state-space structure and prior used to estimate the quantiles (i.e. $\mathbf{F}_t, \mathbf{G}_t, \mathbf{m}_0$, and $\mathbf{C}_0$ discussed in Section 2). We choose to model the quantiles with a second order polynomial trend, which we construct with the `polytrendMod()` function.

```
R> model = polytrendMod(order = 2, m0 = c(mean(LakeHuron),0), C0 = 10*diag(2))
R> model

Prior mean of the state vector:
$m0
         [,1]
```

```
[1,] 579.0041
[2,]   0.0000


Prior covariance of the state vector:
$C0
     [,1] [,2]
[1,]   10    0
[2,]    0   10


Observational vector:
$FF
     [,1]
[1,]    1
[2,]    0


Evolution matrix:
$GG
     [,1] [,2]
[1,]    1    1
[2,]    0    1
```

To allow variability in the dynamic quantile, we select a single discount factor of 0.9 to define the evolution covariance matrix $\mathbf{W}_t$. Discount factors can also be optimized, an example of which we will show in Section 4.2. Further, We place truncated Student-t priors on the skewness parameters via `PriorGamma` as discussed in Section 2.1. For the tracked standard-profile run, we use `n.burn = 2000` and `n.mcmc = 3000`. The R output is shown for only the last quantile using the input `verbose = TRUE`.

```
R> M95 = exdqlmMCMC(y = LakeHuron, p0 = 0.95, model = model,
+                 df = 0.9, dim.df = 2,
+                 PriorGamma = list(m_gam = -1, s_gam = 0.1, df_gam = 1),
+                 n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> M50 = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                  df = 0.9, dim.df = 2,
+                  PriorGamma = list(m_gam = 0, s_gam = 0.1, df_gam = 1),
+                  n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> M5 = exdqlmMCMC(y = LakeHuron, p0 = 0.05, model = model,
+                 df = 0.9, dim.df = 2,
+                 PriorGamma = list(m_gam = 1, s_gam = 0.1, df_gam = 1),
+                 n.burn = 2000, n.mcmc = 3000, verbose = TRUE)
```

Representative console excerpt, abridged to the fields most relevant for the progress format:

```
MCMC backend: C++ (mode=fast)
MCMC init: running LDVB warm start
MCMC start | burn=2000 | keep=3000 | kernel=slice | warm_start=ldvb
...
MCMC done | runtime_sec=122.179 | sigma=0.220 | gamma=5.013 | kernel=slice
```

For this time series of length 98, the tracked standard-profile free-σ run takes about 122.2 seconds to sample 5000 iterations, as seen in the output above. By default `exdqlmMCMC()`

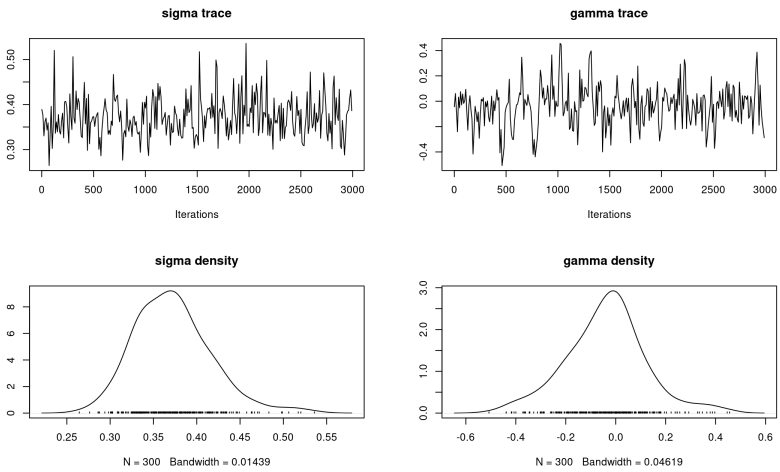

Figure 1: Example 1: Lake Huron. Trace and density plots from the dedicated free-σ median run with 2000 burn-in iterations and 3000 retained draws. For readability, every 10th retained draw is used in the plotting step. The top row shows the posterior trace plots for σ and γ, and the bottom row shows the corresponding posterior densities.

uses the LDVB routine to warm-start the chain, and this warm-start choice can be changed independently of the subsequent MCMC kernel. Under the default setting `mh.proposal = "slice"`, the console output reports compact burn-in and retained-sample progress lines for the current values of σ and γ. For the joint random-walk kernels, the proposal covariance used to sample σ and γ can be supplied manually through `Sig.mh`; alternatively, with the setting `joint.sample = TRUE`, the covariance matrix can be recalculated from the burned sample.

For joint random-walk kernels, the same progress format also reports the current acceptance rate, but this is not applicable under the default slice update. In the targeted support comparison we use to assess this kernel choice for the free-σ Lake Huron median fit, four LDVB-warm-started chains show that the default `slice` update is more stable than `laplace_rw`: the pooled multichain summaries give $\hat{R}$ values of about 1.01 for σ and 1.04 for γ under `slice`, versus about 1.09 and 1.16 under `laplace_rw`. The corresponding effective sample sizes are about 940 and 165 under `slice`, compared with about 129 and 38 under `laplace_rw`, although `laplace_rw` is roughly twice as fast in this support run. We therefore retain `slice` as the default update in this example. The data are relatively symmetric, therefore we do not expect the skewness parameter γ to be far from zero for $p_0 = 0.5$. To examine this, we use a dedicated median run with free σ under the default `slice` update and thin the retained samples by 10 only for plotting. In that run, the posterior mean of σ is about 0.375 with a 95% credible interval $(0.298, 0.473)$, while the posterior mean of γ is -0.044 with a 95% credible interval $(-0.375, 0.321)$. All MCMC samples in the output of `exdqlmMCMC` are `mcmc` objects. Thus we can use the package **coda** to examine the trace and density plots of both σ and γ, seen in Figure 1.

```
R> M50.trace = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                   df = 0.9, dim.df = 2,
+                   PriorGamma = list(m_gam = 0, s_gam = 0.1, df_gam = 1),
```

```
+                   n.burn = 2000, n.mcmc = 3000, verbose = FALSE)
R> keep.idx = seq(1, length(M50.trace$samp.sigma), by = 10)
R> sigma.trace = coda::mcmc(M50.trace$samp.sigma[keep.idx], thin = 10)
R> gamma.trace = coda::mcmc(M50.trace$samp.gamma[keep.idx], thin = 10)
R> coda::traceplot(sigma.trace, main = "sigma trace")
R> coda::densplot(sigma.trace, main = "sigma density")
R> coda::traceplot(gamma.trace, main = "gamma trace")
R> coda::densplot(gamma.trace, main = "gamma density")
```

Clearly γ is indistinct from zero, thus the added flexibility of the skewness parameter is not necessary for the median. We re-run the model with a point-mass prior on γ at 0 using the settings `gam.init = 0` and `fix.gamma = TRUE`, which is equivalent to the setting `dqlm.ind = TRUE`. This reduces the model to the DQLM mentioned in Section 2.5.

```
R> M50 = exdqlmMCMC(y = LakeHuron, p0 = 0.50, model = model,
+                   df = 0.9, dim.df = 2,
+                   gam.init = 0, fix.gamma = TRUE,
+                   n.burn = 2000, n.mcmc = 3000, verbose = TRUE)
```

```
MCMC backend: C++ (mode=fast)
MCMC init: running LDVB warm start
MCMC start | burn=2000 | keep=3000 | kernel=conjugate | warm_start=ldvb
...
MCMC done | runtime_sec=51.838 | sigma=0.430 | kernel=conjugate
```

The results are `exdqlmMCMC` objects that can be used for analysis and forecasting. First we examine the estimated quantiles with the plot method for `exdqlmMCMC` objects, seen in Figure 2. This generates a plot of the data with the MAP estimates and the 95% CrIs of the dynamic quantile. Subsequent quantiles (i.e. $p_0 = 0.50, 0.05$ in this example) can be added to the plot using the function `exdqlmPlot()` with the argument `add = TRUE`.

```
R> plot(M95)
R> exdqlmPlot(M50, add = TRUE, col = "blue")
R> exdqlmPlot(M5, add = TRUE, col = "forest green")
R> legend("topright", lty = 1, col = c("purple","blue","forest green"),
+        legend = c(expression('p'[0]*'=0.95'),expression('p'[0]*'=0.50'),
+                   expression('p'[0]*'=0.05')))
```

Next, we forecast the $k = 8$ step-ahead distributions starting in 1972. To do this, we first create the observational vector (`fFF`) and evolution matrix (`fGG`) that will be used in the forecast updates. In this case both are non-time-varying and identical to those used to estimate the quantile.

```
R> fFF = model$FF
R> fGG =  model$GG
```

We plot the time series data in a narrower range for closer examination and add the forecast estimates using `exdqlmForecast()` with argument `add = TRUE`. Notice we start the forecast at the last time index of the data, i.e. `start.t = length(LakeHuron)`.

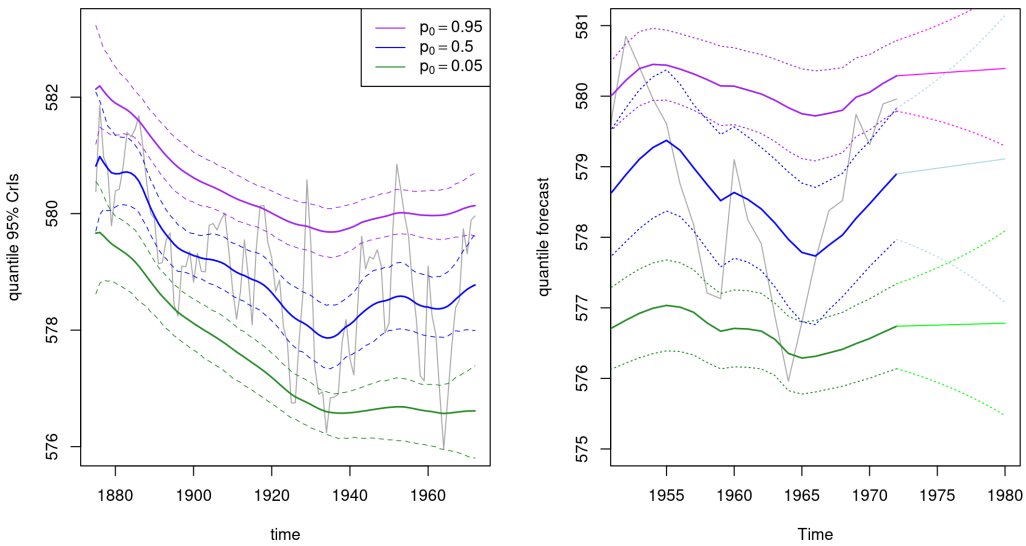

Figure 2: Example 1: Lake Huron. Left: MAP estimates and 95% CrIs of the estimated quantiles, plotted with the data (grey). Right: Forecasted quantile estimates and 95% credible intervals (seen after 1972), along with the filtered quantile estimates and 95% credible intervals (seen from 1952 to 1972).

```
R> plot(LakeHuron,  xlim = c(1952,1980), ylim = c(575,581),
+        col = "dark grey")
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M95,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE)
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M50,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE,
+               cols = c("blue","light blue"))
R> exdqlmForecast(start.t = length(LakeHuron), k = 8, m1 = M5,
+               fFF = fFF, fGG = fGG, plot = TRUE, add = TRUE,
+               cols = c("forest green","green"))
```

This generates a plot of the data with the forecasted quantile estimates and 95% credible intervals (seen after 1972), along with the filtered quantile estimates and 95% credible intervals (before 1972), for reference. Results can be seen in Figure 2. The percentage in the CrIs can be modified with the parameter `cr.percent`.

4.2. Sunspots

For our next example, we will use the yearly Sunspot time series from the package **datasets**. The data are yearly counts for sunspots from 1700 to 1988, shown in Figure 3. With this example we will show how to: use the **dlm** package to create the state-space model, combine blocks of a state-space model, apply the LDVB algorithm, perform visual model diagnostics to compare the exDQLM with the DQLM, and use those diagnostics for model selection.

The solar cycle impacts astronauts in space as well as lives and technology on earth. Of particular interest to scientists is the solar max, or period in which the sun is most active

Fox (2012), therefore we will explore the 0.85 quantile in this example. Again, we begin by building the state-space structure used to estimate the quantile. Here we choose to model the quantile with a first order polynomial trend component as well as a Fourier form seasonal component that incorporates a fundamental period every 11 observations (years), as well as some of its corresponding harmonics. This results in a dynamic model with a total of 9 state parameters, 1 for the first order polynomial component and 8 for the seasonal component. The **exdqlm** functions are compatible with time-invariant `dlm` objects from the **dlm** package. For example, we create the first order trend component with the function `dlmModPoly()` from the **dlm** package, then use `as.exdqlm()` to turn the output into an `exdqlm` object.

```
R> dlm.trend.comp = dlm::dlmModPoly(1, m0 = mean(sunspot.year), C0 = 10)
R> trend.comp = as.exdqlm(dlm.trend.comp)
```

Next, we construct the seasonal component with our `seasMod` function. The duration of sunspot cycles typically follow a period of 11 years, of which we include the first 4 harmonics in the component.

```
R> seas.comp = seasMod(p = 11, h = 1:4, C0 = 10*diag(8))
```

To combine the two components into a single state-space structure, we add the two `exdqlm` objects.

```
R> model = trend.comp + seas.comp
```

The resulting evolution matrix of the combined models is printed for illustration.

```
R> model$GG
```

```
     [,1]    [,2]   [,3]    [,4]   [,5]    [,6]    [,7]    [,8]    [,9]
 [1,]    1  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [2,]    0  0.8413 0.5406  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [3,]    0 -0.5406 0.8413  0.0000 0.0000  0.0000  0.0000  0.0000  0.0000
 [4,]    0  0.0000 0.0000  0.4154 0.9096  0.0000  0.0000  0.0000  0.0000
 [5,]    0  0.0000 0.0000 -0.9096 0.4154  0.0000  0.0000  0.0000  0.0000
 [6,]    0  0.0000 0.0000  0.0000 0.0000 -0.1423  0.9898  0.0000  0.0000
 [7,]    0  0.0000 0.0000  0.0000 0.0000 -0.9898 -0.1423  0.0000  0.0000
 [8,]    0  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000 -0.6549  0.7557
 [9,]    0  0.0000 0.0000  0.0000 0.0000  0.0000  0.0000 -0.7557 -0.6549
```

We will apply discount factors by component, i.e. a discount factor of 0.9 for the trend component of dimension 1 and a discount factor of 0.85 for the seasonal component of dimension 8. This discounting strategy can be applied with arguments `df = c(0.9,0.85)` and `dim.df = c(1,8)`.

We call the function `exdqlmLDVB()` to apply the LDVB algorithm. We use the algorithm with argument `dqlm.ind = TRUE` to estimate the DQLM, as well as with the default settings to estimate the exDQLM. The output is suppressed using input `verbose = FALSE`, however we check the run-time of each model after convergence using the `summary()` generic function. For this dataset of length 289, the LDVB DQLM fit completes in about 9 seconds. The corresponding exDQLM fit is more computationally demanding and takes about 113 seconds.

Updated code/output below to use the current LDVB workflow and current summary output.

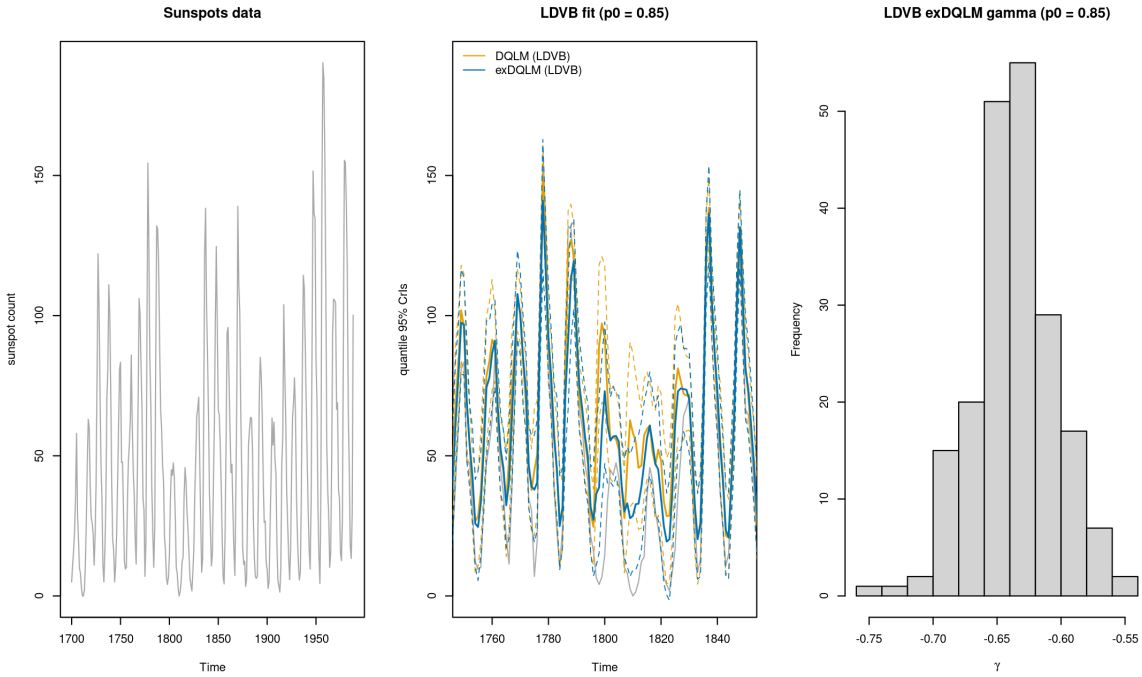

Figure 3: Example 2: Sunspots. Left: The sunspot time series from 1700 to 1988. Center: MAP estimates and 95% CrIs of the estimated quantiles from the DQLM (red) and exDQLM (blue), plotted with the data (grey) from 1750 to 1850. Right: Histogram of samples from the approximated posterior distribution of γ.

```
R> M1 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = c(0.9,0.85), dim.df = c(1,8),
+                  dqlm.ind = TRUE, fix.sigma = FALSE,
+                  verbose = FALSE)
R> M2 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = c(0.9,0.85), dim.df = c(1,8),
+                  fix.sigma = FALSE,
+                  verbose = FALSE)
R> summary(M1)

Bayesian Dynamic Quantile Regression Model (DQLM)
Number of Observations: 289
State Dimension: 9
Discount factors ( dimensions ): 0.9 ( 1 ), 0.85 ( 8 )

DQLM fitted using VB
Iterations until convergence: 67
Run-time: 9.128 seconds

R> summary(M2)

Bayesian Dynamic Quantile Regression Model (exDQLM)
Number of Observations: 289
State Dimension: 9
Discount factors ( dimensions ): 0.9 ( 1 ), 0.85 ( 8 )
```

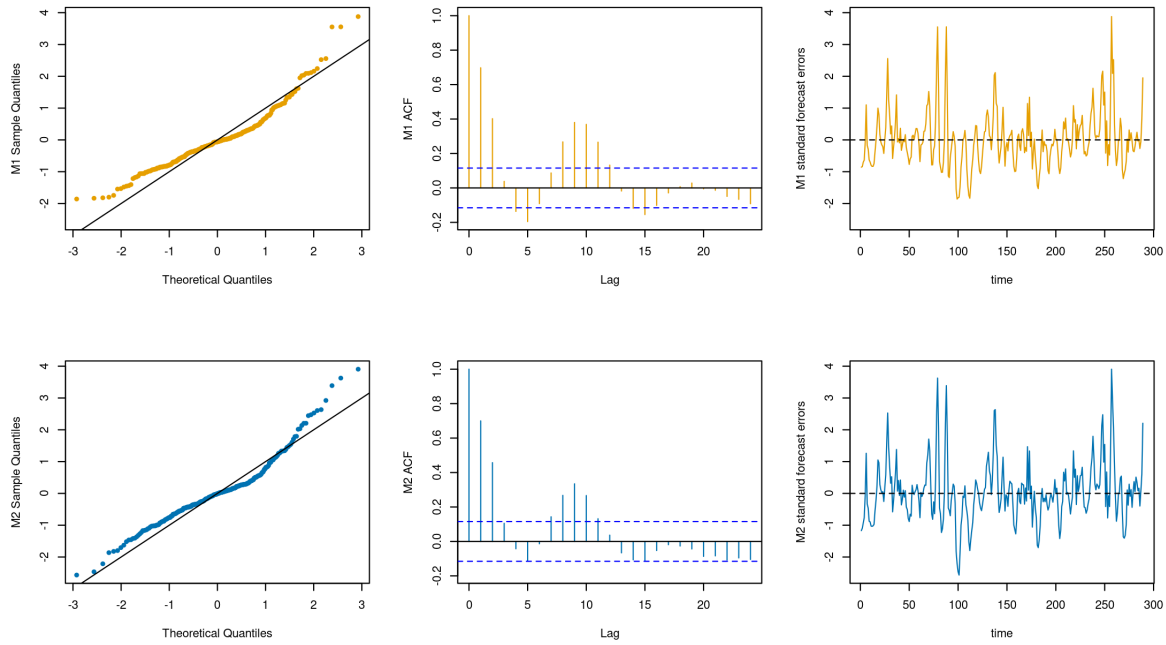

Figure 4: Example 2: Sunspots. The QQ-plots (left column), ACF plots (center column) and standard forecast errors (right column) from the DQLM (red) and exDQLM (blue).

```
exDQLM fitted using LDVB
Parameters estimated with LD: gamma sigma
Iterations until convergence: 200
Run-time: 113.952 seconds
```

We can visualize the differences between the two resulting dynamic quantiles using the function `exdqlmPlot()`, seen in Figure 3. For clarity we limit the figure to years 1750 to 1850.

```
R> plot(sunspot.year, xlim = c(1750,1850), col = "dark grey",
+       ylab = "quantile 95% CrIs")
R> exdqlmPlot(M1, add = TRUE, col = "red")
R> exdqlmPlot(M2, add = TRUE, col = "blue")
```

To examine whether the added flexibility of the exDQLM is significant, we can also visualize the approximate posterior distribution of the skewness parameter γ by plotting the samples from the corresponding variational distribution, also seen in Figure 3.

```
R> hist(M2$samp.gamma,xlab=expression(gamma),main="")
```

The approximate distribution is clearly distinct from zero, thus we expect a better model fit from the exDQLM.

To further examine the two models, we use the function `exdqlmDiagnostics()` to plot the model diagnostics discussed in Section 3. The results are seen in Figure 4. The QQ-plot and the ACF of the standard forecast errors both favor the exDQLM.

```
R> exdqlmDiagnostics(M1, M2, cols = c("red","blue"))
```

Finally, the function `exdqlmDiagnostics` can also be used for model selection. Say we want to check whether the discount factor we used for the seasonal component was optimal with respect to the KL divergence of the one-step-ahead predictive distributions. We can create a list of possible discount factors, quickly run the LDVB for each possible model, and select the set of discount factors which results in the lowest KL divergence. A simple example is seen below, which takes approximately 6 minutes to run.

```
R> possible.dfs = cbind(0.9,seq(0.85,1,0.05))
R> possible.dfs
```

```
     [,1] [,2]
[1,]  0.9 0.85
[2,]  0.9 0.90
[3,]  0.9 0.95
[4,]  0.9 1.00
```

```
R> KLs <- vector("numeric")
R> ref.samp = rnorm(length(sunspot.year))
R> for(i in 1:nrow(possible.dfs)){
+   temp.M2 = exdqlmLDVB(y = sunspot.year, p0 = 0.85, model = model,
+                  df = possible.dfs[i,], dim.df = c(1,8),
+                  sig.init = 2, verbose = FALSE)
+   temp.check = exdqlmDiagnostics(temp.M2, plot = FALSE,
+                              ref = ref.samp)
+   KLs = c(KLs,temp.check$m1.KL)
+ }
R> # optimal dfs based off KL divergence
R> possible.dfs[which.min(KLs),]
```

```
[1] 0.90 0.85
```

The results suggest the discount factor we used for the seasonal component is optimal.

4.3. Big Tree water flows

For our final example, we consider the average monthly natural water flow (cubic feet per second) at the Big Tree gauge of the San Lorenzo river in Santa Cruz County, CA from January 1937 through April 2021 (U.S. Geological Survey 2016), for a total of 1012 observations. The dataset is included in our package as `BTflow`. With this example we show how to build a state-space structure manually (particularly for a dynamic quantile regression), apply our transfer function exDQLM, examine individual components of the estimated quantiles, and perform quantitative model diagnostics to compare models.

For this particular dataset, the interest is in low quantiles as a possible indicator of drought, thus we will consider the $p_0 = 0.15$ quantile. To investigate any climatological influence on this low quantile, we include the Niño 3.4 index (Rayner, Parker, Horton, Folland, Alexander, Rowell, Kent, and Kaplan 2003) as in input of our exDQLM. The Niño 3.4 index is a measure of how anomalously warm or cool the average sea surface temperature is in the Niño 3.4 region (5°S-5°N, 170°W-120°W) of the Pacific Ocean. It is one of the most commonly used indices

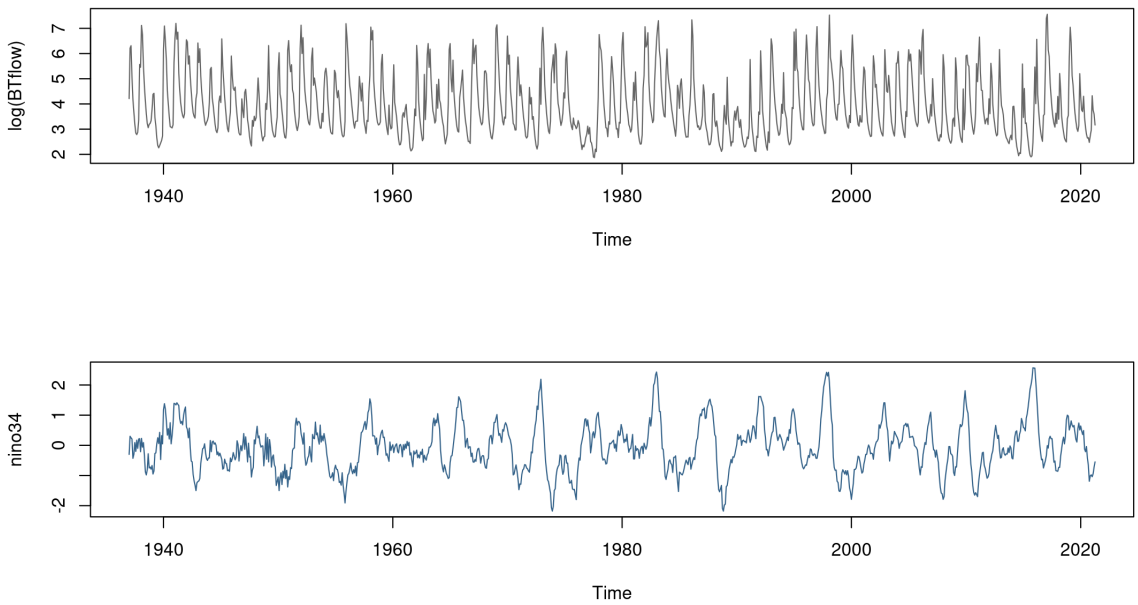

Figure 5: Example 3: Big Tree water flow. Top: Average monthly water flow at the Big Tree gauge of the San Lorenzo river in Santa Cruz County, CA, plotted on the log-scale. Bottom: Monthly values of Niño 3.4. Both time series span January 1937 through April 2021.

to define El Niño and La Niña events. The monthly Niño 3.4 dataset used in our analysis is also included in the package as `nino34`. We model the relationship between Niño 3.4 and the quantile with two model structures: a dynamic quantile regression and a transfer function as seen in Section 2.4. To avoid modeling problems caused by the scale and extremes in the Big Tree dataset, we fit the following models to the time series on the log scale seen in Figure 5.

In addition to the components that capture the relationship with the Niño 3.4 index, we also include a first order polynomial trend component as well as a Fourier form seasonal component that includes a single periodic component with period 12. The Big Tree water flow time series follows a period of twelve months, of which we include only the first (annual) harmonic.

```
R> trend.comp = polytrendMod(1, m0 = 3, C0 = 0.1)
R> seas.comp = seasMod(p = 12, h = 1, C0 = diag(1,2))
R> model = trend.comp + seas.comp
```

Next, we create the dynamic regression component for our first model using `regMod()`. Combining this with the trend and seasonal component we have our complete state-space structure for the dynamic quantile regression model.

```
R> reg.comp = regMod(nino34, m0 = 1, C0 = 1)
R> model.w.reg = model + reg.comp
```

To allow the Niño 3.4 input components of both models to vary over time, we set the respective discount factors to be 0.95 (this is done via the parameter `df` in `exdqlmLDVB()` and the

parameter `tf.df` in `transfn_exdqlmLDVB()`). Alternatively, we set the discount factors to 1 and 0.9 for the trend and seasonal components, respectively. This allows the seasonal amplitudes to evolve over time, while any further variation in the 0.15 quantile from year to year will solely be attributed to the effects of the Niño 3.4 input in our models.

Updated paragraph. The transfer function model is created within the function `transfn_exdqlmLDVB()` and follows the structure detailed in Section 2.4. On the coarse candidate grid used in this illustration, optimization using the KL divergence suggests the effects of the Niño 3.4 index decay at an optimal transfer function rate of $\lambda = 0.85$.

```
R> possible.lams = c(0.1,0.3,0.5,0.7,0.85)
R> KLs <- vector("numeric")
R> ref.samp = rnorm(length(BTflow))
R> for(i in 1:length(possible.lams)){
R>   temp.M2 = transfn_exdqlmLDVB(y = log(BTflow), p0 = 0.15, model = model,
+                               df = c(1,0.9), dim.df = c(1,2),
+                               X = nino34, tf.df = c(0.95), lam = possible.lams[i],
+                               tf.m0 = c(0,0), tf.C0 = diag(c(0.1,0.005),2),
+                               sig.init = 0.1, gam.init = - 0.1,
+                               tol = 0.05, verbose = FALSE)
R>   temp.check = exdqlmDiagnostics(temp.M2, plot = FALSE,
+                             ref = ref.samp)
R>   KLs = c(KLs,temp.check$m1.KL)
R> }
R> # optimal lambda based off KL divergence
R> possible.lams[which.min(KLs)]
```

```
[1] 0.85
```

With these discount factors and optimal λ, we estimate the dynamic quantile regression model (`M1`) and transfer function model (`M2`).

```
R> M1 = exdqlmLDVB(y = log(BTflow), p0 = 0.15, model = model.w.reg,
+                df = c(1,0.9,0.95), dim.df = c(1,2,1),
+                sig.init = 0.1, gam.init = - 0.1,
+                tol = 0.05, verbose = FALSE)
R> M2 = transfn_exdqlmLDVB(y = log(BTflow), p0 = 0.15, model = model,
+                        df = c(1,0.9), dim.df = c(1,2),
+                        X = nino34, tf.df = c(0.95), lam = 0.85,
+                        tf.m0 = c(0,0), tf.C0 = diag(c(0.1,0.005),2),
+                        sig.init = 0.1, gam.init = - 0.1,
+                        tol = 0.05, verbose = FALSE)
```

To visualize the differences between the two resulting dynamic quantiles we use the function `exdqlmPlot`, results of which are found in Figure 6. For clarity we limit the figure to view 1970 to 1990.

```
R> plot(log(BTflow), col = "grey", ylim = c(1,8), xlim=c(1970,1990),
+       ylab = "quantile 95% CrIs")
R> exdqlmPlot(M1, add = TRUE)
R> exdqlmPlot(M2, add = TRUE, col = "forest green")
```

Clearly the resulting quantiles have significant differences. To identify from where these differences arise, we can utilize the function `compPlot` to examine the components of the

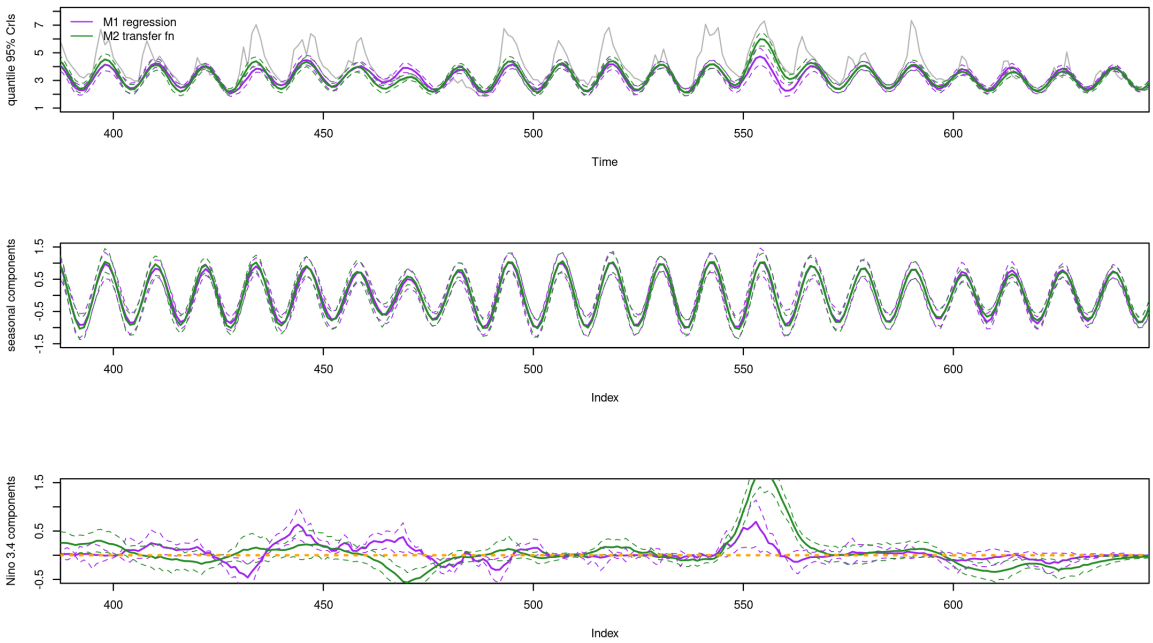

Figure 6: Example 3: Big Tree water flow. Top: MAP estimates and 95% CrIs of the estimated quantiles from the models with the regression component (purple) and transfer function component (green), plotted with the data (grey) from 1970 to 1990. Middle: MAP estimates and 95% CrIs of the annual seasonal components. Bottom: MAP estimates and 95% CrIs of the components which model the quantile association with Niño 3.4. The estimates of the dynamic regression component $X_t\boldsymbol{\theta}_{4,t}$ of M_1 where $\boldsymbol{\theta}_{4,t}$ denotes the fourth element of the state vector, are seen in purple. The estimates of the transfer function component ζ_t of M_2 are in green. The horizontal orange dotted line is at zero for reference.

models more closely. We examine the annual component of the parameter vector first. The annual component is comprised of the second and third elements of the parameter vector $\boldsymbol{\theta}_t$, thus we set the parameter `index = c(2,3)` to visualize this component. The result is seen in Figure 6, again limited to the years 1970 to 1990.

```
R> plot(NA, ylim = c(-1.5,1.5), xlim = c(1970,1990),
+         ylab = "seasonal components")
R> compPlot(M1, index = c(2,3), add = TRUE)
R> compPlot(M2, index = c(2,3), add = TRUE, col = "forest green")
```

Although the seasonal components of the two models exhibit differences in amplitude, we expect more significant differences to appear in the Niño 3.4 component of each model. Again, we use the function `compPlot` to visualize these components seen in Figure 6 from 1970 to 1990.

```
R> plot(NA, ylim = c(-0.5,1.5), xlim = c(1970,1990),
+          ylab = "NiÃśo 3.4 components")
R> compPlot(M1, index = c(4), add = TRUE)
R> compPlot(M2, index = c(4,5), add = TRUE, col = "forest green")
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
```

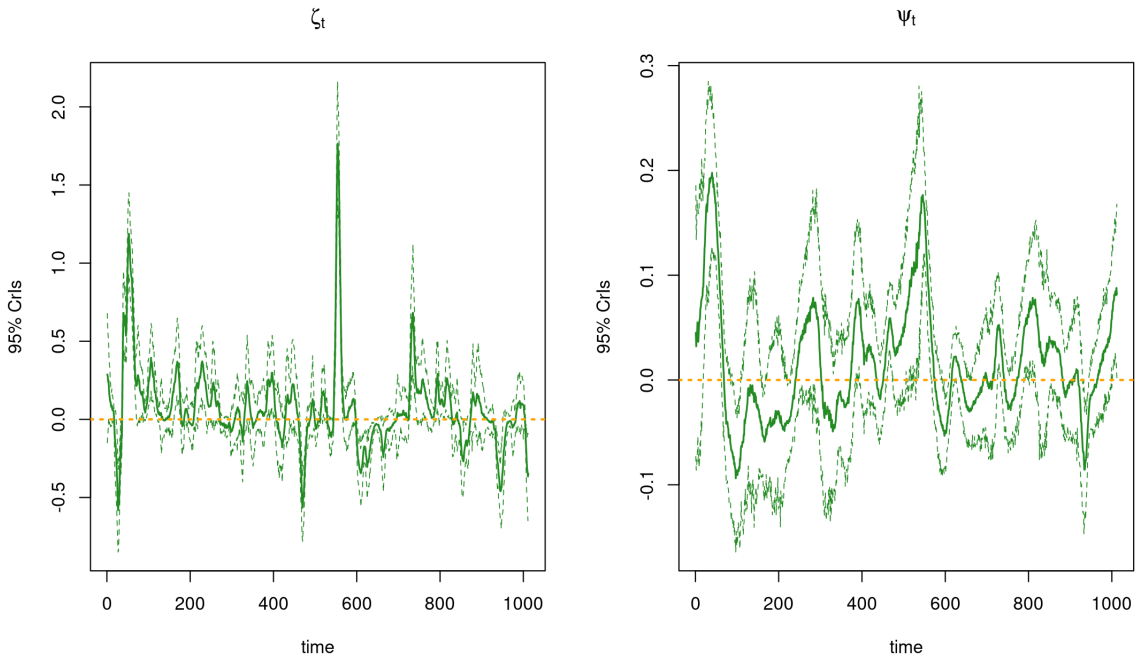

Figure 7: Example 3: Big Tree water flow. MAP estimates and 95% CrIs of parameters ζ_t and ψ_t from the transfer function component. Horizontal dotted orange lines are at zero for reference.

For the transfer function model, `M2`, it is also of interest to look at the estimates of parameters ζ_t and ψ_t directly. Because these parameters augmented the state-space model as discussed in Section 2.4, the function `compPlot` can also be used to look at these estimates directly using the parameter `just.theta = TRUE`. This setting returns the dynamic posterior estimates of a single index of the state vector $\boldsymbol{\theta}_t$ (or $\tilde{\boldsymbol{\theta}}_t$ in the case of the transfer function model). Parameter `index` of `4` and `5` correspond to ζ_t and ψ_t, respectively. Results can be found in Figure 7.

```
R> compPlot(M2, index = 4, col= "forest green",
+             add = FALSE, just.theta = TRUE)
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
R> title(expression(eta[t]))
R> compPlot(M2, index = 5, col = "forest green",
+             add = FALSE, just.theta = TRUE)
R> abline(h = 0, col = "orange", lty = 3, lwd = 2)
R> title(expression(psi[t]))
```

To further understand these transfer function parameters, we examine the median number of time steps until the effect of Niño 3.4 on the 0.15 quantile is less than or equal to 1e−3, as discussed in Section 2.4. A median value 22.47 suggests the effect of the Niño 3.4 index value at time t lasts for around 22 months.

```
R> M2$median.kt

[1] 22.46999
```

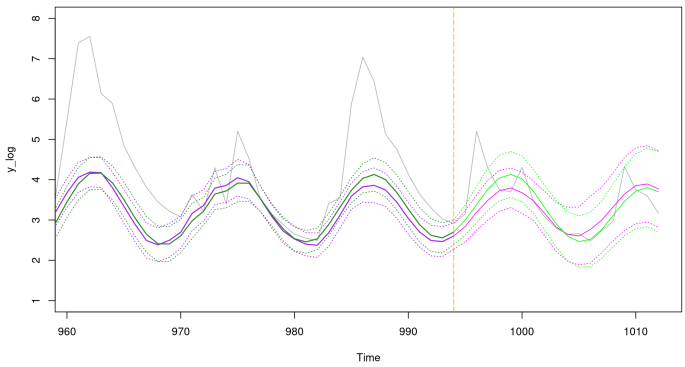

Figure 8: Example 3: Big Tree water flow. 18-step-ahead quantile forecast beginning October 2019 through April 2021 overlaid on the log Big Tree water flow time series. The vertical dot-dashed line is at the beginning of the forecast period, October 2019, for reference. Solid lines indicate mean estimates and dotted lines 95% CrI, with the filtered estimates seen leading up to October 2019 and the forecast estimates beyond October 2019. Estimates from M_1, the quantile regression model, are seen in purple/pink. Estimates from M_2, the transfer function model, are seen in green.

	M1		M2	
	output	value	output	value
pplc	`m1.pplc`	306.41	`m2.pplc`	312.04
KL	`m1.kl`	0.257	`m2.kl`	0.224

Table 6: Example 3: Big Tree water flow. Model diagnostic output from exdqlmDiagnostics() and resulting values.

This suggests a long-term forecast could be of interest. Thus we forecast the $k = 18$ step ahead distributions for both models starting in October 2019, results of which are seen in Figure 8.

```
R> exdqlmForecast(start.t = length(BTflow)-18, k = 18, m1 = M1)
R> exdqlmForecast(start.t = length(BTflow)-18, k = 18, m1 = M2,
+                    add = TRUE, cols = c("forest green","green"))
```

Finally, we use the function `exdqlmDiagnostics` to compute the KL divergence and pplc discussed in Section 3, results of which are seen in Table 6.

```
R> checks.out = exdqlmDiagnostics(M1, M2,
+                                 cols = c("purple","orange"), plot = FALSE)
```

The refreshed diagnostics are not unanimous. The KL divergence favors the nonlinear relationship captured by the transfer function model, whereas the pplc is slightly smaller for the regression model. In this example we therefore treat the transfer function specification as a useful alternative whose advantages are most visible in the fitted quantile paths and transfer-function components, rather than as a uniformly dominant model under every diagnostic criterion.

4.4. Static exAL regression on a simulated heteroskedastic example

This subsection is new.

To illustrate the static routines under sparse shrinkage, we consider a correlated Gaussian regression benchmark for which the target conditional quantile is known exactly. Let

$$\mathbf{x}_i \sim N_8(\mathbf{0}, \Sigma_x), \qquad (\Sigma_x)_{jk} = 0.5^{|j-k|},$$

with sparse signal

$$\boldsymbol{\beta}^\star = (3,\ 1.5,\ 0,\ 0,\ 2,\ 0,\ 0,\ 0)^\top.$$

New benchmark setup. For each target quantile level p_0, we generate

$$Y_i^{(p_0)} = \mathbf{x}_i^\top \boldsymbol{\beta}^\star + \sigma_\varepsilon\{Z_i - \Phi^{-1}(p_0)\}, \qquad Z_i \sim N(0,1), \qquad \sigma_\varepsilon = 1.5.$$

Then

$$Q_Y(p_0 \mid \mathbf{x}_i) = \mathbf{x}_i^\top \boldsymbol{\beta}^\star,$$

so the true p_0-quantile is the same sparse linear signal at each fitted quantile level. We use $n = 160$ training observations and an independent holdout sample of size 800, and fit separate static exAL models at $p_0 \in \{0.05, 0.25, 0.50\}$. Both fits use the regularized horseshoe (RHS) prior of Nishimura and Suchard (2023), implemented in the package through `beta_prior = "rhs_ns"`, with $\tau_0 = 0.15$, fixed slab $\zeta^2 = 9$, and an unshrunk intercept. We fit `exal_static_LDVB()` and `exal_static_mcmc()`; the MCMC routine uses its default `slice` kernel and is warm-started from the LDVB fit. The lower-tail LDVB fit at $p_0 = 0.05$ converges more slowly, so we allow a larger LDVB iteration budget there. For reproducibility, we fix the dataset seed at `20260714`, selected in a preliminary screening exercise that required both methods to recover the active signals, LDVB to remain faster than MCMC at each fitted quantile, and the LDVB holdout-loss and coefficient-recovery metrics to stay close to the MCMC baseline. Setting `dqlm.ind = TRUE` would recover the AL special case discussed in Section 2, but here we focus on the general static exAL model under sparse RHS shrinkage. Updated code block below to add the new static sparse benchmark.

```
R> Sigma.x = 0.5 ^ as.matrix(dist(1:8))
R> beta.true = c(3, 1.5, 0, 0, 2, 0, 0, 0)
R> rhs.ctrl = list(tau0 = 0.15, a_zeta = 2, b_zeta = 9,
                   zeta2_fixed = 9, shrink_intercept = FALSE)
R> set.seed(20260714)
R> sim.y = function(X.raw, p0) {
      drop(X.raw %*% beta.true) + 1.5 * (rnorm(nrow(X.raw)) - qnorm(p0))
    }
R> X.raw = MASS::mvrnorm(160, mu = rep(0, 8), Sigma = Sigma.x)
R> X = cbind(1, X.raw)
R> p.grid = c(0.05, 0.25, 0.50)
R> y.train = lapply(p.grid, function(p0) sim.y(X.raw, p0))
R> M.ldvb = lapply(p.grid, function(p0) exal_static_LDVB(
+                    y = y.train[[which(p.grid == p0)]], X = X, p0 = p0,
+                    beta_prior = "rhs_ns",
+                    beta_prior_controls = rhs.ctrl,
+                    max_iter = ifelse(p0 == 0.05, 420, 260),
+                    tol = 1e-4, verbose = FALSE))
R> M.mcmc = lapply(p.grid, function(p0) exal_static_mcmc(
+                    y = y.train[[which(p.grid == p0)]], X = X, p0 = p0,
```

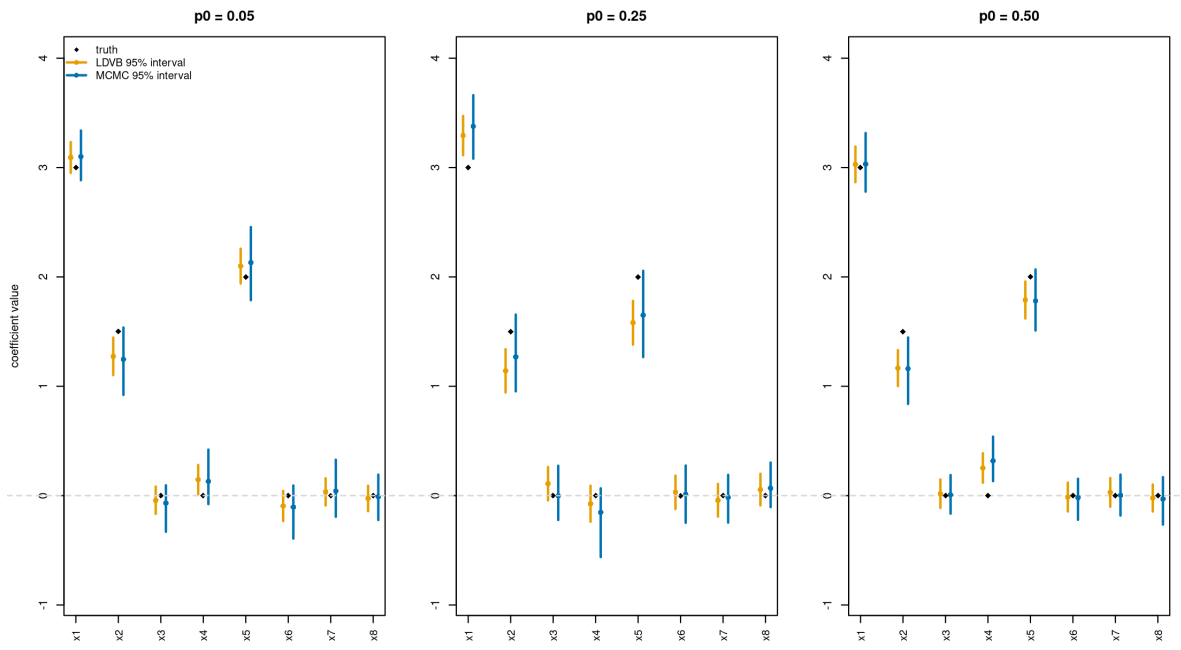

Figure 9: Example 4: sparse static exAL regression under the correlated Gaussian benchmark with the RHS prior. Black diamonds denote the true non-intercept coefficients, orange intervals denote LDVB 95% posterior intervals, and blue intervals denote MCMC 95% posterior intervals. Panels correspond to p0 = 0.05, 0.25, and 0.50.

```
+                    beta_prior = "rhs_ns",
+                    beta_prior_controls = rhs.ctrl,
+                    n.burn = 1000, n.mcmc = 300,
+                    init.from.vb = TRUE,
+                    verbose = FALSE))
```

New static diagnostics paragraph. For static fits, the helper function `exalDiagnostics()` summarizes fitted quantiles on a common design matrix and, when a reference quantile is supplied, reports truth-recovery metrics directly. For example, on an independent holdout design:

```
R> X.hold.raw = MASS::mvrnorm(800, mu = rep(0, 8), Sigma = Sigma.x)
R> X.hold = cbind(1, X.hold.raw)
R> ref.hold = drop(X.hold %*% c(0, beta.true))
R> y.hold = sim.y(X.hold.raw, 0.25)
R> exalDiagnostics(M.ldvb[[2]], M.mcmc[[2]],
+                  X = X.hold, y = y.hold, ref = ref.hold,
+                  plot = FALSE)
```

Updated figure interpretation. Figure 9 shows that both methods recover the active coefficients and shrink the null coefficients toward zero across the lower tail, lower-middle quantile, and median. The lower-tail fit at $p_0 = 0.05$ is the most demanding case, yet both methods recover the sparse signal cleanly; on this screened run, LDVB is slightly sharper on the active coefficients, whereas MCMC is slightly better on the holdout quantile RMSE and null

p_0	method	runtime (s)	active RMSE	null MAE	holdout quantile RMSE
0.05	LDVB	18.99	0.152	0.069	0.473
0.05	MCMC	39.46	0.163	0.048	0.452
0.25	LDVB	15.46	0.231	0.070	0.374
0.25	MCMC	40.25	0.235	0.060	0.370
0.50	LDVB	6.68	0.414	0.046	0.504
0.50	MCMC	40.42	0.349	0.046	0.445

Table 7: Example 4: runtime and sparse-signal recovery metrics for the static exAL fits under the RHS prior. The active RMSE is computed over the three nonzero coefficients, the null MAE is computed over the five zero coefficients, and the holdout quantile RMSE is computed against the true target quantile on an independent holdout sample of size 800.

shrinkage. At $p_0 = 0.25$, the two fits remain closely aligned, with LDVB marginally better on the active-signal metric and MCMC slightly better on the holdout and null-shrinkage metrics. At $p_0 = 0.50$, both fits again track the true sparse pattern, but MCMC is modestly stronger on the active-signal and holdout metrics, while the null-shrinkage results are nearly indistinguishable.

Updated table discussion. Table 7 reinforces the visual comparison. The two methods are competitive across all three quantile levels, but the relative advantage is not uniform: MCMC is modestly stronger on the holdout-recovery and null-shrinkage metrics at $p_0 = 0.05$, the two methods are nearly tied at $p_0 = 0.25$, and MCMC is modestly stronger on the active-signal and holdout metrics at $p_0 = 0.50$. LDVB, however, is faster in all three fits for the selected data set, with runtime reductions of about 2.1-fold at $p_0 = 0.05$, 2.6-fold at $p_0 = 0.25$, and 6.1-fold at $p_0 = 0.50$. In this benchmark, static LDVB offers meaningful computational savings as a first-pass approximation while still recovering the sparse signal well, but this speed comes with the usual variational tradeoff that posterior uncertainty quantification may be less faithful than the exact MCMC baseline. Accordingly, the static RHS specification supports both roles cleanly: LDVB for rapid screening and model exploration, and static MCMC as the reference method when higher-fidelity posterior uncertainty quantification is required.

5. Conclusion

In this paper we presented the R package **exdqlm** for Bayesian estimation and analysis of flexible dynamic quantile linear models. The package implements exact posterior inference via MCMC and fast approximate posterior inference via LDVB, and it also includes routines for transfer function exDQLMs, static exAL quantile regression, forecasting, and model diagnostics. Together, these functions provide a practical framework for dynamic and static quantile modeling beyond the AL special case while preserving a computationally convenient latent-variable representation.

The examples illustrate the main workflows supported by the package. The dynamic examples show how **exdqlm** can be used to construct state-space models, estimate dynamic quantiles, compare the exDQLM with its AL special case, examine transfer function effects, and evaluate model fit through diagnostics and forecasting. The static example shows how

the same framework can be used for sparse quantile regression with both LDVB and MCMC, with LDVB serving as a fast first-pass approximation and MCMC providing the reference analysis when more faithful posterior uncertainty quantification is required.

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Journal of Statistical Software http://www.jstatsoft.org/
published by the Foundation for Open Access Statistics http://www.foastat.org/
MMMMMM YYYY, Volume VV, Issue II *Submitted:* yyyy-mm-dd
doi:10.18637/jss.v000.i00 *Accepted:* yyyy-mm-dd